

Kaixiang (Lucas) Zhuang

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EDUCATION

Carnegie Mellon University

Aug. 2026 – Present

M.S. in Software Engineering | Part-Time Program for Working Professionals

University of California, Irvine

June 2026

B.S. in Computer Science; B.S. in Business Information Management

Intelligent Systems (AI) | GPA: 3.92

EXPERIENCE

Founder & Software Engineer, AI Products

Apr. 2026 – Present

Blux Inc. (AI4Science)

Irvine, CA

- Designed an AI-native research workflow platform for biological researchers, translating complex scientific workflows into interactive software systems.
- Built backend services and agent-orchestration pipelines connecting AI reasoning with research data, workflow state, and human approval.
- Architected data-ingestion, retrieval, and workflow-execution layers for traceable AI-assisted research workflows.

Software Engineer Intern

Jun. 2025 – Sep. 2025

Yahoo Finance

Mountain View, CA

- Built an LLM-powered AI analyst for stock valuation and earnings analysis using AWS Bedrock, Lambda, and SQS, enabling users to run financial valuation workflows in under two seconds through natural-language prompts.
- Developed retrieval infrastructure for 6,500+ public companies using PostgreSQL and JSONB, enabling low-latency access to financial data and generated analyses.
- Reduced third-party API usage by 40% and request latency by 200 ms through Redis caching and bulk data ingestion.
- Productionized event-driven AI inference services using FastAPI, Docker, Kubernetes (EKS), Terraform, Lambda, and SQS to support reliable long-running workflows.

Software Engineer Intern

Jan. 2024 – Mar. 2024

University of California, Irvine

Irvine, CA

- Led a team of three in building a distributed search engine spanning web crawling, indexing, ranking, storage, and query serving.
- Architected an N-gram inverted index containing 1.88M MongoDB records, achieving sub-second search retrieval.
- Built a web crawler indexing 370,000+ URLs across 55,000+ websites with trap detection.

PROJECTS AND RESEARCH

Undergraduate Researcher

Sep. 2024 – Jun. 2026

University of California, Irvine

Irvine, CA

- Built multi-agent planning systems in C++ and Python for warehouse-scale coordination, implementing Conflict-Based Search and benchmarking infrastructure.
- Reduced total planning time complexity from exponential to linear through imitation learning, enabling coordination across thousands of agents.

Fabflix.Life (Distributed E-Commerce Platform) [Demo] | *Node.js, Express.js, AWS, Kubernetes, MySQL*

- Built a distributed e-commerce platform with Node.js, Express.js, and MySQL, implementing REST APIs and secure transactional workflows.
- Measured a 44% performance improvement and 11,137 requests per minute in Apache JMeter load tests after deploying with Kubernetes (EKS), Elastic Load Balancing, and Auto Scaling.
- Designed high-concurrency data access using MySQL connection pooling.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, C, SQL, Shell/Bash

Backend & AI: FastAPI, Node.js, Express.js, LangChain, AWS Bedrock, RAG, MCP, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud & Infrastructure: AWS Lambda, SQS, EKS, Docker, Kubernetes, Terraform